Ysbyty Gwynedd
Anaesthetics Department
Adran Anaestheteg

PER-OPERATIVE

YG CG 410 Nov 2023

Perioperative Anaesthetic Management for Robotic Surgery

FIRST APPROVED: NOVEMBER 2023 REVIEW BY: NOVEMBER 2026

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Key Points

- Anaesthetic risk is largely associated with positioning, restricted access to patient & increased length of surgery.
- Available evidence demonstrates that robotic surgery confers a degree of benefit over traditional laparoscopic surgery for high BMI patients in terms of length of hospital stay.
- Most frequent complications are haemodynamic changes, altered pulmonary function, raised intracranial and intraocular pressures (hence cerebral oedema), peripheral neuropathies, corneal abrasions, limb compartment syndrome, and airway oedema.

Aim

- To provide guidance to those infrequently managing Robotic Surgery lists.

Pre-Operative

- Routine pre-op assessment focusing on assessing respiratory and cardiovascular status in anticipation for prolonged pneumoperitoneum and steep Trendelenburg (30°).
- Relative contraindications include:
  - Intolerance of steep Trendelenburg (consider open surgery in the context of significant cardiorespiratory compromise)
  - Glaucoma
- Group and Save

Intra-Operative

Consider combined Spinal analgesia/GA technique. D/W Surgeon, the likelihood of open conversion. Based on local experience, spinal opioids, tend not to be needed for Gynae cases.

The key issues to consider are the:

- Effect of pneumoperitoneum & steep Trendelenburg on respiratory mechanics, cardiovascular stability and cerebral/airway oedema.
- Limited access to patient during surgery.
- Implications of poor patient positioning.

Intervention	Rationale
Induce on trolley in AR	To avoid having to slide patient down table already on a beanbag. Final robot checklist will take place in theatre prior to transferring patient from anaesthetic room.
ETT	Check for Endobronchial intubation on 'Head down'. Before extubation, deflate cuff to check for leak to assess airway oedema
Check ventilation parameters are acceptable in head down position	Check patient tolerates head down position on trolley in AR. Once the Robot is in position, you will not be able to reduce head down position once set.
Muscle Paralysis	After robot docking, patient movement can cause harm. Consider the use of muscle relaxant infusion. Use quantitative neuromuscular monitoring throughout. Reverse relaxant after the robot has been undocked from the patient.
Invasive BP monitoring +/- vasopressor infusion	To manage hypotension whilst ensuring relative fluid restriction. Reduces risk of orbital/cerebral/airway and surgical site oedema. Have a low threshold for siting arterial monitoring and setting up vasopressor infusion in the context of difficult intraoperative access.
Consider Oro-Gastric Tube	To aid gastric drainage and avoid pooling of gastric secretions on face. Stomach acid reflux can cause severe oral ulceration and conjunctival burns.
Large bore IV access with long extensions	Minimal access to patient. PVC +/- A-line should be sited on side to be tilted up. This is to avoid patient sliding onto arm increasing extravasation risk.
Temp probe	Long surgery and limited options for warming.
Meticulous Eye Protection	Increased risk of corneal abrasions due to conjunctival oedema & incomplete closure of eyelid. Close eyelids and protect with clear pads. Consider eye ointment. Suction mouth/NGT before tilting head down.
Analgesia	LA infiltration of port sites IV Paracetamol +/- NSAID. Medium-long-acting Opioid towards the end. Consider the use of magnesium/clonidine as a multimodal approach.
VTE Prophylaxis	TEDS

Positioning:

Meticulous care should be taken to minimise risk associated with prolonged surgery in steep head down position.

- Supine or Lloyd Davies – on a bean bag or non-slip memory foam. No draw sheets to reduce risk of pressure areas.
- Head in a neutral position - consider gel head ring or placing the pillow in 'portrait position to allow beanbag to be moulded better into the shoulders.

- Ensure beanbag is moulded tightly around shoulders.
- Arms wrapped and slight flexion at elbows (reducing ulnar injury)
- Hands supported by gel pads in neutral position.
- Make sure monitoring cables are not stuck between bean bag and patient.
- Supine only: Pillow under knees to relieve lower back tension & Heal pads

Once the patient is put in the steep Trendelenburg position, then patient position should be re-checked ensuring that:

- Head remains neutral
- No endobronchial intubation
- No pressure/traction on the arms and that the hands are in a neutral position.
- Arterial line still working
- Heal pads in correct place, if used.

TIVA Considerations

TIVA is well suited for long cases. However surgeons may require both arms by sides, rather than one arm on chest (Gynae) and the TIVA cannula will not be visible or accessible. If TIVA is used, it should be administered through a large bore cannula placed in the arm on the side to be tilted upwards with fluid running to maintain patency.

If there are any concerns the patient should be switched to volatiles or if hard indication for TIVA consider central line access instead.

Post-Operative

- Analgesia - routine post-op analgesia +/- PCA.
- Antiemetic
- Laxatives

Cardiac Arrest

- Robotic Surgical Instruments must be disengaged prior to any movement of the operating table or patient. This will likely delay initiation of chest compressions by around 60 seconds.
- This should ideally be simulated as a drill with any staff unfamiliar with robotic surgery
- DC Shock can be delivered with robot docked, if necessary.

References

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